文献汇报

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文献

Huanjie Tao,Xiaobo Lu.**Smoke vehicle** detection based on robust **codebook** model and robust volume **local binary count** patterns[J]. Image and Vision Computing,2019,86.

车辆尾烟检测

- 场景：交通监控视频中检测车辆尾烟
- 挑战：车辆的干扰、烟雾的变化、照明、阴影。
- 目的：降低误报率
- 方法：
  - R-Codebook模型：检测可疑烟雾块
  - Completed VLBC：时空特征
  - 改进CVLBC得到R-VLBC：在抗噪性和获取各像素的信息之间取得平衡；获取多尺度信息

Related work

Table 1
The used features in smoke detection literatures.

Feature kinds		Features and literatures
Spectral features		Image graininess, intensity values and histogram [1], Mean, variance, skewness and kurtosis [2,43], Smoke color filtering [3], chromatic configuration [43]
		Color moment (CM) [5], Assessing smoke probability [6],
		Intensity and saturation density in the HSI color space [7], rule-based chrominance detection [28]
		Rule and threshold based color features [4,8,17,18,39–42],
Spatial features	Texture features,	Local binary patterns (LBP) [7,14,15,34,47], LBP pyramids, LBP variance (LBPV) pyramids [9], Multi-scale block Tamura (MSB-Tamura) [10], Pairwise comparing LBP (PCLBP) [13],
		Non-redundant LBP (NRLBP) [16]. Variable Orientation GLCM (V-GLCM) [3]
		High-order Local Ternary Patterns (HLTP) [12] Hamming-distance-based LBP (HDLBP) [46];
		LBP co-occurrence with HDLBP (LBPCoHDLBP) [46]
		Gray level co-occurrence matrix (GLCM) [11,20,32,34],
		HLTP based on magnitudes of noise removed derivatives and values of center pixels (HLTPMC) [12];
	Gradient-based features	Shape features [3,6,17], Contour [2],
		Edge orientation histogram (EOH) [7], Hu moments [2,7].
		Histogram of oriented gradient (HOG) [14,18,43]; Haar-like [7,19];
		Range filtering on three orthogonal planes (RF-TOP) [5];
		Surface roughness and area randomness [41]
	Spatial learning-based features	Dictionary learning [22,23]. Convolutional neural networks (CNN) [24,27];
		Deep multi-scale CNN (DMCNN) [25]; Deep normalization and CNN (DNCNN) [26]
		Holistic learning-based high-order feature [48]
	Spatial frequency domain features	Discrete wavelet transform (DWT) and wavelet [4,20,21,28,34,43],
		Discrete cosine transform (DCT) [21,22].
Temporal/ spatiotemporal features	Motion and optical flow	Histogram of oriented optical flow (HOOF) [18], Spatiotemporal bag-of-features [18]
		Motion orientation histogram (MOH) [5], Motion orientation model [40],
		Optical smoke flow pattern using Gabor filter [8], motion vector [41,43]
		Optical flow [42]; transmission [44],
		Trajectories, right-leaning line, smooth streamline, low-frequency, source and vertical–horizontal ratio [31]
		Motion energy and saliency map [45].
		Sharp edge; growing region; rising region detection, perimeter disorder [39]
	Dynamic texture	Volume LBP (VLBP) [29], LBP on three orthogonal planes (LBPTOP) [29],
		Complete VLBP (CVLBP) [29], swaying and diffusion model based on GLCM [32],
		Non-redundant local motion binary patterns (NRLMBP) [16]
		Map of brightness differences, edge map, and Laplacian map in Spatio-Temporal LBP (STLBP) [30]
		Weber local descriptors in three orthogonal planes (WLD-TOP) [33],
	Spatiotemporal math model	Surfacelet transform and HMT model [34,49], higher order linear dynamical systems (hLDS) [35],
	Spatiotemporal learning-based features	Spatial–temporal ConvNet [36], recurrent convolutional network [37],
		Multi-scale and multi-order features from 3D local differences [38].
	Spatiotemporal frequency domain features	Spatial–temporal wavelet [33], spatial–temporal energy using DWT [8]

检测可疑烟雾块：R-Codebook

- CodeBook算法能得到每个像素的时间序列模型。

  CB={CW1,CW2,…CWn,t}

  CW={lHigh,lLow,max,min,t_last,stale}

- 遍历像素对应的背景模型CB中的码字CW，若存在一个CW，使其满足一定阈值，则该像素为背景，否则为前景。
- Codebook在复杂环境中的速度很慢。为了实现运动对象的实时和鲁棒性检测，提出R-Codebook。
- 改进：使用策略（不为每个像素构造码本）加速算法；用LBP特征提高对照明变化、相机晃动、树叶晃动的鲁棒性。

检测可疑烟雾块

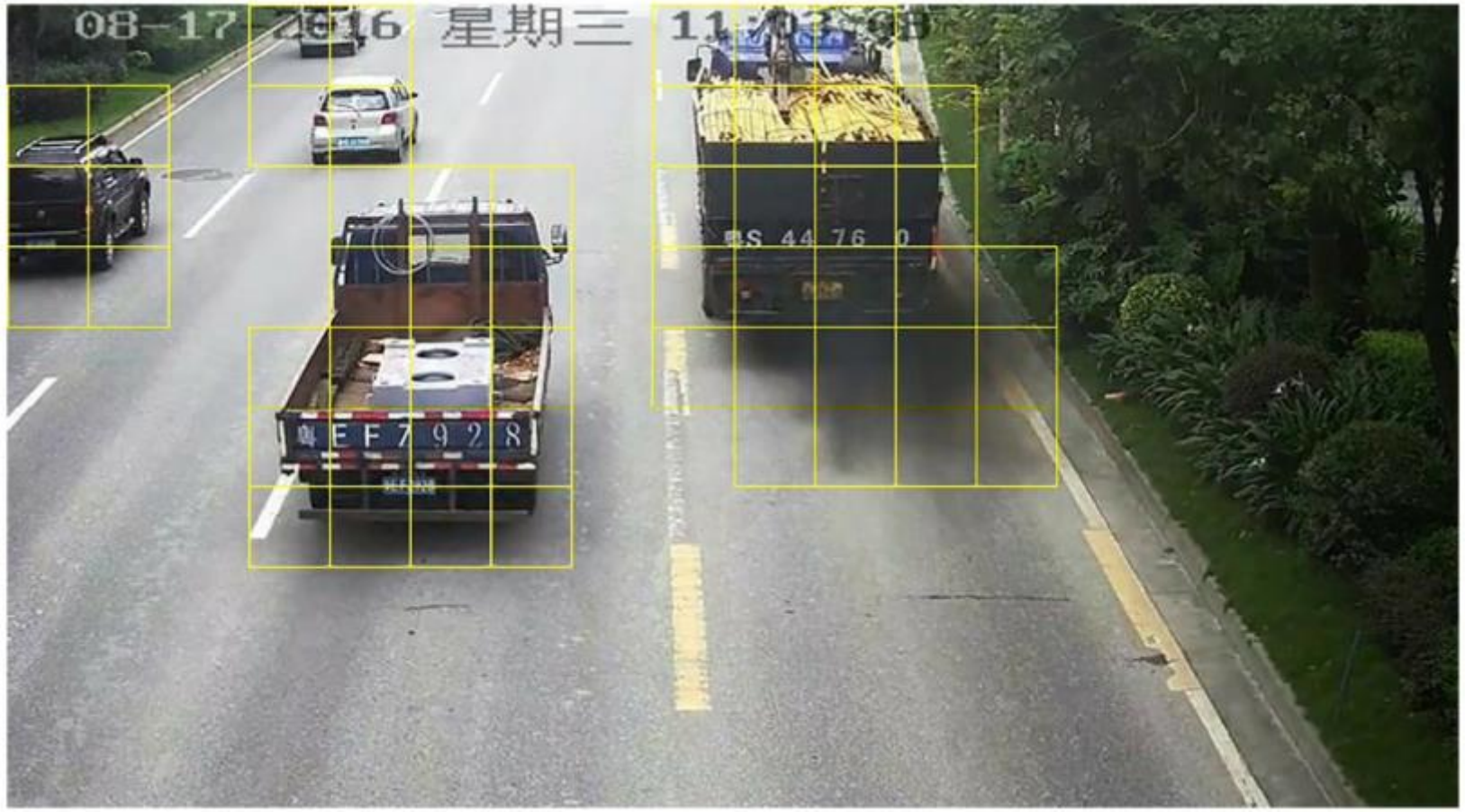

Fig. 2. Suspected smoke blocks (yellow blocks) using R-Codebook model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

LBC

- LBC用于2D灰度纹理分类。

$$LBC_{P,R} = \sum_{i=0}^{P-1} s(g_i(R) - g_c) \qquad s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

- 为实现灰度值缩放的不变性，仅保留符号并计算正号的数量。
- LBP描述符专注于局部结构特征，但是LBC仅考虑邻域内有多少像素比中心像素强度值高。
- LBC对照明变化有鲁棒性。

VLBC

- 将2D纹理LBC扩展到时空域形成3D VLBC。

$$V^{(3P+3)} = \{g_{t_c-L,c}, g_{t_c-L,0}, \cdots, g_{t_c-L,P-1}, g_{t_c,c}, g_{t_c,0}, \cdots, g_{t_c,P-1}, g_{t_c+L,0}, \cdots, g_{t_c+L,P-1}, g_{t_c+L,c}\}$$

$$VLBC_{L,P,R} = \sum_{p=0}^{3P+1} s(v_p - g_{t_c,c})$$

$$s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

- VLBC可以描述宏观结构，记录中心像素与其相邻像素之间的差异来表征时空信息。

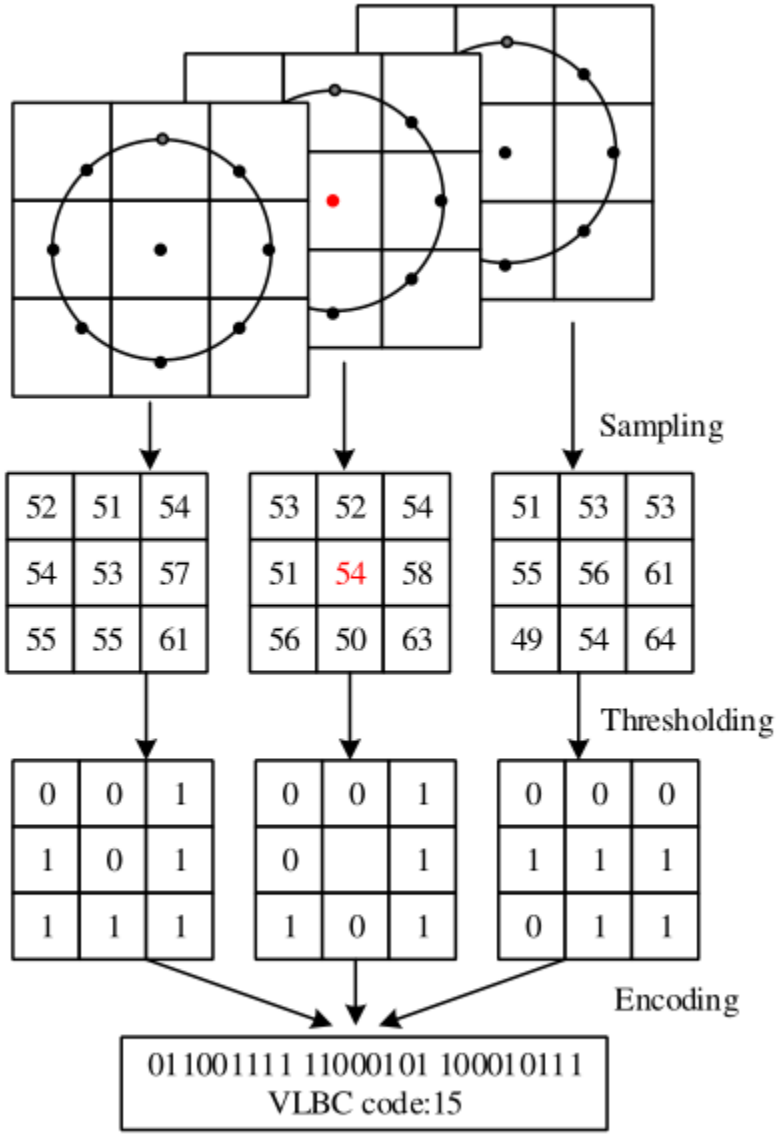

Fig. 3. An illustration to compute $VLBC_{1,8,1}$.

减小噪声：WLT

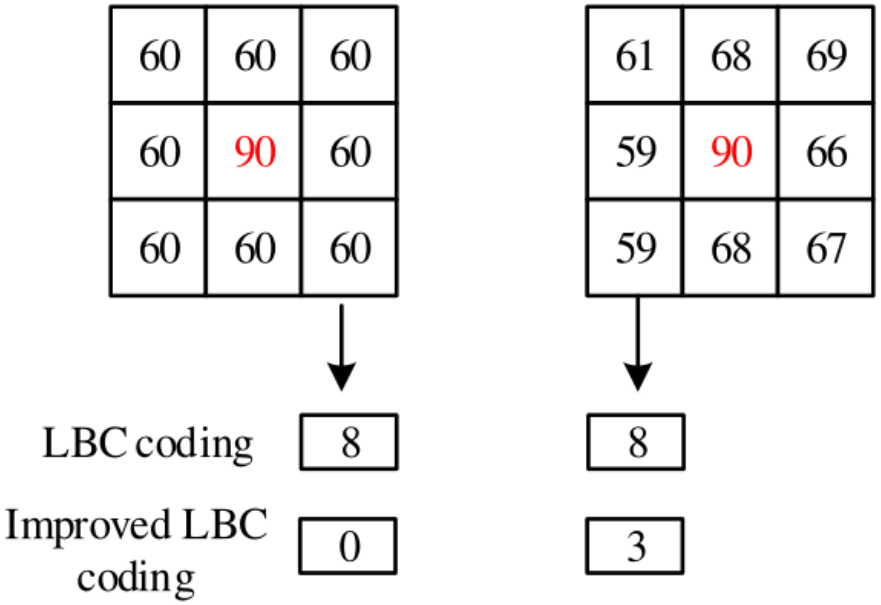

Fig. 4. Different patterns may have the same LBC code.

- 为了**减小噪声**的影响，将中心像素的值替换为WLT。

$$WLT = \frac{\alpha \sum_{p=0}^{3P+1} v_p + g_{t_c,c}}{\alpha(3P+2)+1}$$

- α为1，则WLT等于平均灰度值；α为0，则WLT等于中心像素。参数α用于平衡**抗噪性**和获取各个像素的**信息**。

NR-VLBC

- 定义NR-VLBC如下

$$NR-VLBC_{L,P,R} = \min\{OVLBC_{L,P,R}, 3P+2-OVLBC_{L,P,R}\}$$

$$OVLBP_{L,P,R} = \sum_{p=0}^{3P+1} h(u_p - WLT)$$

$$h(u_p - WLT) = \begin{cases} 1, & \text{if } u_{p-1} - WLT > 0 \\ h(u_{p-1} - WLT), & \text{if } u_{p-1} - WLT = 0 \\ 0, & \text{if } u_{p-1} - WLT < 0 \end{cases}$$

- NR-VLBC反映前背景之间的对比度。？
- 原始VLBC代码及其补码被视为相同情况特征向量的维数将减少一半。

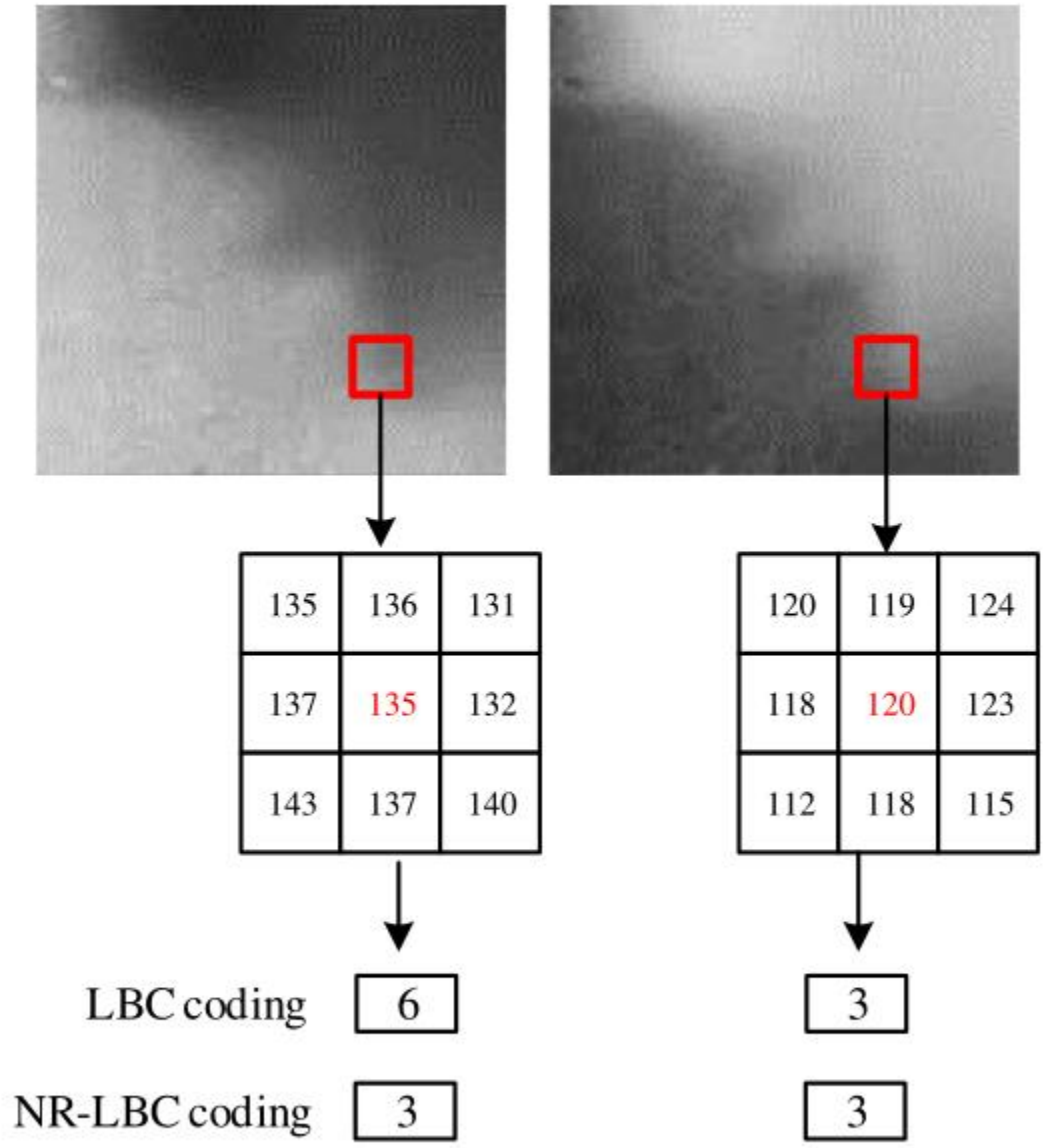

Fig. 5. Two images represent the same smoke structure with inverted background and foreground.

CVLBC

- VLBC可以对局部差异分布进行建模，但是它忽略了灰度差异的大小，并丢失了局部对比度信息。构造complete VLBC。

$$CVLBC = \{VLBC_S, VLBC_M, VLBC_C, VLBC_D\}$$

- VLBC_S等效于NR-VLBC；
- VLBC_M表示幅值差别；
- VLBC_C表示局部灰度等级；$VLBC_C_{L,P,R} = s\left(g_{t_c,c} - c_l\right)$
- VLBC_D表示邻域像素的梯度信息。$VLBC_D_{L,P,R} = \sum_{p=0}^{3P/2} s\left(u_i - u_{i+3P/2+1}\right)$

135	132	131
137	135	132
143	137	140

(a)

0	-3	-4
2		-3
8	2	5

(b)

-1	-1	-1
1		-1
1	1	1

(c)

0	3	4
2		3
8	2	5

(d)

-5	-5	-12
		-5

(e)

Fig. 6. (a) A local structure. (b) The local differences. (c) The sign. (d) The magnitude components. (e) The central symmetry difference.

级联特征

Two ways, including jointly or hybridly, can be used to combine the four operators. In this paper, we first build two 2-D joint histograms, "VLBC_S/D" and "VLBC_M/C". Then the histograms are converted to two 1D histograms. We concatenate the two 1D histograms to generate a joint histogram denoted by "VLBC_S/D_M/C",

$$CVLBC_{S/D_M/C} = \{H_{S/D}, H_{M/C}\}$$

$$H_{S/D}(s,d) = \sum^{(3P/2+1)^2-1} f_1(CVLBC(p), s, d)$$

$$f_1(R{-}VLBC, s, d) = \begin{cases} 1, & if \;\; VLBC_S(p) = s, \\ & VLBC_D(p) = d \\ 0, & otherwise \end{cases}$$

$$H_{M/C}(m,c) = \sum_{p=0}^{2(3P+2)-1} f_2(CVLBC(p), m, c)$$

$$f_2(R{-}VLBC, m, c) = \begin{cases} 1, & if \;\; VLBC_S(p) = m, \\ & VLBC_D(p) = c \\ 0, & otherwise \end{cases}$$

Multi-scale features

- 由于距相机不同距离的烟雾将呈现不同的比例，将多尺度策略引入CVLBC，以形成R-VLBC。
- 通过改变P和R来获得多尺度信息。（ P领域的取点数目，R是邻域半径）

$$R-VLBC=\left\{CVLBC_{S/D_M/C}^{(s1)},CVLBC_{S/D_M/C}^{(s2)},CVLBC_{S/D_M/C}^{(s3)}\right\}$$

- 其中s=(R,P)∈{（1,8），（2,16），（3,24）}。

检测流程

- 用RCodebook模型提取可疑烟雾块，在可疑块不同帧的相同位置提取其前块和后块形成块序列。从块序列中提取R-VLBC特征，用SVM进行分类。
- 块序列的分类结果作为当前块的分类结果，如果当前块为烟雾块，则其对应的帧将被分类为烟雾帧。在连续的100帧中，如果超过8个是烟雾帧，则当前视频将被分类为烟雾视频。

数据集

Table 2
All the used datasets.

Datasets	Descriptions
Dataset 1	Contains 11,034 smoke block sequences with heavy smoke vehicles
Dataset 2	Contains 6618 smoke block sequences with light smoke vehicle
Dataset 3	Contains 1,341,890 non-smoke block sequences without shadows
Dataset 4	Contains 124,377 non-smoke block sequences with shadows
Dataset 5	Contains 5937 smoke frames and 151,613 non-smoke frames

块序列分类性能

Table 3
Comparison experiments of different methods on block sequences classification.

Different methods	DR of smoke block sequences		FAR of non-smoke block sequences	
	Dataset 1	Dataset 2	Dataset 3	Dataset 4
Method in [3]	0.9603	0.7120	0.1520	0.2689
Method in [6]	0.9631	0.7135	0.1690	0.2985
Method in [7]	0.9634	0.7200	0.1844	0.2843
Method in [8]	0.9572	**0.7396**	0.1427	**0.2442**
Method in [40]	0.9514	0.6734	0.2344	0.3481
Method in [9]	0.9536	0.6884	0.2031	0.3035
Method in [12]	0.9581	0.6955	0.1879	0.2912
Method in [13]	0.9612	0.7013	0.1912	0.3044
Method in [14]	0.9588	0.7024	0.1887	0.2966
Method in [15]	0.9556	0.6993	0.1988	0.3132
Method in [18]	0.9577	0.7212	0.1436	0.2899
Method in [25]	**0.9667**	0.7243	**0.1320**	0.2961
Our method	**0.9733**	**0.8116**	**0.0930**	**0.1473**

$$DR = \frac{TP}{P}$$

$$FAR = \frac{FN}{N}$$

帧分类性能

Table 4
Experimental results of performance comparisons with other methods on frames classification.

Different methods	Dataset 5	
	DR	FAR
Method in [3]	0.8854	0.1331
Method in [6]	0.8958	0.1716
Method in [7]	0.8963	**0.1423**
Method in [8]	0.8701	0.1256
Method in [40]	0.8407	0.2301
Method in [9]	0.8452	0.1803
Method in [12]	0.8602	0.1610
Method in [13]	0.8871	0.1834
Method in [14]	0.8733	0.1703
Method in [15]	0.8550	0.2010
Method in [18]	0.8790	0.1433
Method in [25]	**0.9022**	0.1755
Our method	**0.9253**	**0.0893**

$$DR = \frac{TP}{P}$$

$$FAR = \frac{FN}{N}$$

参数α选择

$$DR = \frac{TP}{P}$$

$$FAR = \frac{FN}{N}$$

Table 5
Experiments results of using R-VLBC with different parameter α on block sequences classification.

R-VLBC with different α	DR of smoke block sequences		FAR of non-smoke block sequences	
	Dataset 1	Dataset 2	Dataset 3	Dataset 4
R-VLBC ($\alpha = 0$)	0.9617	0.8001	0.1044	0.1598
R-VLBC ($\alpha = 1/74$)	0.9710	0.8103	0.0950	0.1493
R-VLBC ($\alpha = 1/50$)	0.9733	**0.8116**	**0.0930**	0.1473
R-VLBC ($\alpha = 1/26$)	0.9735	0.8110	0.0968	**0.1456**
R-VLBC ($\alpha = 1$)	**0.9744**	0.8106	0.0972	0.1489

NR-VLBC

$$DR = \frac{TP}{P}$$

$$FAR = \frac{FN}{N}$$

Table 6
Experimental results of R-VLBC with and without non-redundant property on frames classification.

Different methods	Dataset 5	
	DR	FAR
R-VLBC without non-redundant property	0.9041	0.1082
R-VLBC with non-redundant property	**0.9253**	**0.0893**

CVLBC性能

$$DR = \frac{TP}{P}$$

$$FAR = \frac{FN}{N}$$

Table 7
Experimental results of performance comparisons with other methods on frames classification.

Different methods	Dataset 5	
	DR	FAR
VLBC_S/D	0.9014	0.1211
VLBC_M/C	0.8957	0.1243
VLBC_S/D_M/C	**0.9253**	**0.0893**

多尺度性能

$$DR = \frac{TP}{P}$$

$$FAR = \frac{FN}{N}$$

Table 8
Experimental results of R-VLBC with different scales on frame sequences classification.

Different scales	Dataset 5	
	DR	FAR
R-VLBC with single scale 1	0.8997	0.1157
R-VLBC with single scales 2	0.9023	0.1097
R-VLBC with single scale 3	0.9012	0.1057
R-VLBC with multi scale	**0.9253**	**0.0893**

多尺度性能

$$DR = \frac{TP}{P}$$

$$FAR = \frac{FN}{N}$$

Table 9
Experiments to verify the advantages of R-Codebook model on frame classification.

Different methods	Dataset 5	
	DR	FAR
Original Codebook model	0.9094	0.1153
R-Codebook model	**0.9253**	**0.0893**

多尺度性能

Fig. 8. Some smoke frames captured from surveillance videos. The red blocks denote the correctly recognized smoke blocks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

总结

- 提出基于R-Codebook和R-VLBC的车辆尾烟检测方法，以降低误报率。
- 用R-Codebook检测可疑烟雾块，然后用R-VLBC识别烟雾块序列。
  - 组合VLBC_S，VLBC_M，VLBC_C和VLBC_D来提出CVLBC，以表征时空特征。
  - 基于两种策略进一步改进得到R-VLBC。一种是通过WLT代替中心像素来克服对噪声敏感的问题。另一个是通过改变邻域半径和采样点数目来识别烟雾块序列，以提取多尺度信息。

Thank You!